700-km intercity Yb/Sr optical clock comparison at 2.3×10^{-18}

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Abstract

Redefining the International System of Units (SI) second requires independent frequency-ratio measurements between clocks based on different atomic transitions in different laboratories, with an overall uncertainty below 5×10^{-18} and consistent results. However, the self-assessed uncertainties of these clocks have not yet been validated by such comparisons. We compared a ytterbium optical lattice clock in Wuhan with a strontium optical lattice clock in Shanghai—about 700 km apart—over a phase-stabilized fiber link spanning 1 350 km, and determine the Yb/Sr frequency ratio to be 1.207 507 039 343 337 720 1(28), with a Bayesian statistical uncertainty of 1.14×10^{-18} and a total uncertainty of 2.34×10^{-18}. The value differs from the NIST–JILA result 1.207 507 039 343 337 723 0(37) by -2.87×10^{-18}, consistent within the combined standard uncertainty. To our knowledge, this is the first intercity frequency-ratio measurement to meet the uncertainty requirement of the roadmap for the redefinition of the SI second, and together with the NIST–JILA result it provides the second, independent determination of the Yb/Sr ratio by a different institute, as required by criterion I.2.b for this pair of transitions. The measurement also demonstrates, at intercity scale, the capability that a wide-area optical-clock network requires.

1 Introduction

Optical lattice and single-ion clocks now reach fractional systematic uncertainties at or below 2×10^{-18} [1–8], two orders of magnitude better than current primary frequency standards[9;10]. Comparing independent clocks in different regions and laboratories verifies these self-assessed precision budgets more objectively: the environmental conditions and geodetic assumptions entering a self-evaluation are typically tied to a single location, and identical methods within one laboratory may share common systematic biases. Linking clocks of different types and locations into a collaborative network is therefore a key pathway toward establishing optical clocks as a verifiable metrological infrastructure[11;12]. The idea has moved from proposal to practice: fiber links with noise levels below those of the clocks themselves now span thousands of kilometers[13;14], and multi-clock campaigns have measured tens of ratios among clocks in six countries[15] and across a continental network[16]. Satellite-based advanced links with comparable performance[17–20] are also being developed and will interconnect with fiber networks to form a future global optical clock network.

The most immediate purpose of such a network is the redefinition of the SI second. One of the mandatory requirements (I.2.b) in the roadmap for this redefinition[21] is that at least five non-unit frequency ratios between standards of different clock transitions be measured, each ratio at least twice by different institutes, in agreement with an overall uncertainty of the comparison $\Delta\nu/\nu < 5 \times 10^{-18}$—a requirement reiterated in the 2025 CCTF progress report[22;23]. However, comparisons across independent institutions and different locations have reached at best a precision of 7.7×10^{-18} [16]. So far, no frequency-ratio measurements between clocks in different cities have satisfied this uncertainty requirement[24].

Closing that gap matters beyond the definition itself. Chronometrically leveled clocks map the gravitational potential of the Earth at centimeter resolution[1;25;26], and clock networks have been proposed and used as sensors for dark matter[27–29]. All of these applications assume that remote clocks agree at a level that, until now, has been demonstrated only locally, in three



Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilized 1 550 nm fiber link. Relay stations along the route carry erbium-doped fiber amplifiers and fiber-noise-cancellation loops.

experiments: a Sr/Yb/Al^+ triple-clock ratio measurement with an uncertainty of 3.2×10^{-18} by NIST–JILA in Boulder (US)[30], and two transportable-clock comparisons at the 10^{-18} level[26;31].

Here we report an intercity frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km apart, connected by a 1 350 km fiber link. We determine the Yb/Sr ratio with a total uncertainty of 2.34×10^{-18}; gravitational redshift effects have been corrected, whereas measurement error and the time-varying gravitational redshift term associated with gravitational tides have been retained as uncertainty components. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and demonstrates, at intercity scale, the class of ratio measurement that the redefinition process depends on.

2 Results

2.1 Network and observation campaign

Figure 1 shows the two-clock network. A ytterbium (Yb) lattice clock at IAPMST[8] in Wuhan and a strontium (Sr) lattice clock at USTC[5] in Shanghai are separated by 700 km and compared through a 1 350 km phase-stabilized fiber link (2 700 km loop-back) on a commercial telecom network used non-exclusively, with the carrier transmitted from Shanghai to Wuhan on the ITU-T 100 GHz DWDM channel C34 (1 550.12 nm, 193.4 THz). The link extends the field-deployed link of ref.[14], using the same cascaded fiber-noise-cancellation technique across 21 independent relay stations, each carrying an erbium-doped fiber amplifier (22 EDFAs in total); its added instability stays below the clock noise for averaging times beyond 100 s.

Both combs are self-referenced and phase-locked to the local clock signal, and all other frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations (698/1397 nm on the Sr side; 578/1156 nm on the Yb side), so the two clocks are the system's only independent frequency sources. The recorded quantity is the out-of-loop beat between the received carrier and the local Yb-referenced comb, sampled every second and converted to fractional units by normalizing to the carrier frequency f_{1550} (1 550.12 nm); the fiber loop-back is monitored out of loop throughout. The geopotential difference between the two clocks was

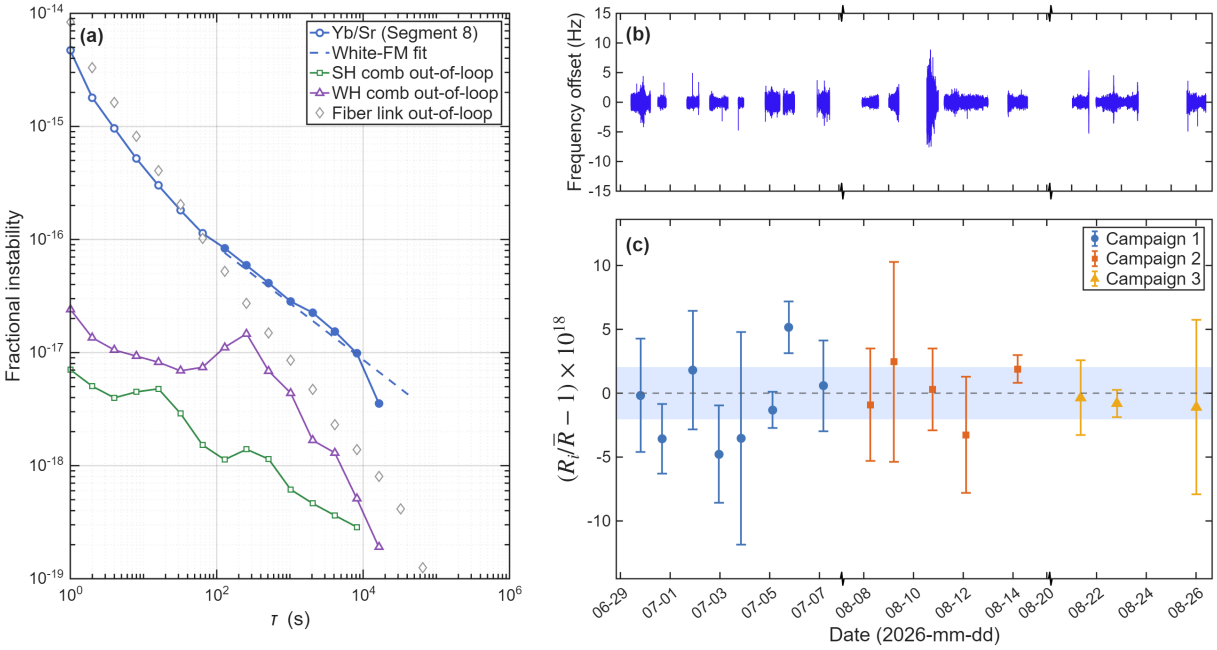


Figure 2: Frequency-ratio measurement results. (a) Fractional instability of a representative segment (Yb/Sr, segment 8); the gray markers are the out-of-loop instabilities of the two frequency combs (SH, WH) and of the fiber link. (b) Per-segment ratio deviations from the final reported value. (c) The same deviations by campaign date.

measured by spirit leveling to be $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$ (see Methods). Each observation window (segment) yields a frequency-ratio estimate through the fixed frequency-chain relations (see Methods).

2.2 Clock-ratio determination

Over two months (58 days) the two clocks and their transfer chains ran synchronously for a total of 274 effective hours, recorded as 16 jump-free segments; several of the segment boundaries correspond to deliberate clock shutdowns and restarts, so the reproducibility of the ratio rests on tested restart behavior rather than on continuous operation. The segment estimates show no systematic drift over the campaign, and the network tolerates interruption, provided each retained window is self-consistently characterized. The results are presented in Figure 2.

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty (Methods). Following recent ratio comparisons[16;30], we combine the segments with a random-effect model that admits a common excess scatter[32;33], and evaluate three estimators (Table 1). The three agree at the 10^{-18} level; their center values spread by $4.4 \times 10^{-}21$. We adopt the Bayesian random-effect estimate, which carries the largest standard uncertainty of the three, as the primary value,

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1, \tag{1}$$

with statistical uncertainty 1.139×10^{-18}.

2.3 Uncertainty budget

Beyond the statistical uncertainty of the ratio combination, the comparison carries several systematic contributions: the published evaluations of the two clocks, the geodetic determination of the gravitational-potential difference between the sites, and the residuals of the transfer subsystems. We discuss each in turn and combine them into the total standard uncertainty (Table 2).

Table 1: Statistical combination of the 16 segment ratios. WLS denotes weighted least squares with inverse-variance weights $w_i = 1/u_i^2$; ξ is the inferred common excess scatter. χ^2_{red} is the reduced chi-square of the WLS combination (15 degrees of freedom, $p = 9.8 \times 10^{-1}$, Birge ratio 0.642).

Estimator	Yb/Sr	u (10^{-18})	ξ (10^{-18})	χ^2_{red}
WLS (inverse variance)	1.207 507 039 343 337 720 130	1.128	—	0.412
Mandel–Paule	1.207 507 039 343 337 720 130	1.128	0.00	1
Bayesian random effect	1.207 507 039 343 337 720 134	1.139	0.14	—

Table 2: Uncertainty budget of the Yb/Sr frequency-ratio measurement. Components are treated as independent.

Component	Contribution (10^{-18})
Statistical (Bayesian random effect)	1.139
Sr systematics	0.92
Yb systematics	1.3
Static gravitational potential	0.946
Fiber link	0.1
Frequency comb	0.1
Time-varying redshift	0.85
Total (quadrature)	2.34

The Sr clock contributes 0.92×10^{-18} from its published systematic evaluation[5], and the Yb clock 1.3×10^{-18}[8]. The static gravitational-potential term is the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$ converted to fractional frequency as $u(\Delta W)/c^2$, giving 0.946×10^{-18}; this is the same leveled geopotential difference already applied as the static correction (Section 2.1). The frequency conversion stages each contribute 0.1×10^{-18} and 0.1×10^{-18} from the frequency-reference settings of the frequency conversion.

The time-varying gravitational redshift is more involved. Solid Earth and ocean tides, together with atmospheric, hydrological and non-tidal ocean loading, move the local potential by amounts reaching the 10^{-18} level over the campaign[34–36]. Correcting the beat record with a tide template from the public software ETideLoad4.5[37], which implements the IERS Conventions (2010)[38], degrades rather than improves the fit, raising the reduced chi-square of the combination from 0.41 to 0.54, and a second, independent template built from the DE421 ephemerides[39] and the FES2022 ocean-tide model[40] does the same ($\chi^2_{\mathrm{red}} = 0.50$) while moving the center in the opposite direction, by -0.85×10^{-18} versus $+0.21 \times 10^{-18}$ (Methods). The sign disagreement between the two models confirms that the templates are incomplete at this level, so the time-varying term is carried as an uncertainty rather than a correction, at the larger of the two model shifts, 0.85×10^{-18} (Methods).

Combining the statistical and systematic components in quadrature gives the total standard uncertainty 2.34×10^{-18}; the reported ratio is therefore

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1(28), \tag{2}$$

where the parenthesized digits are the total standard uncertainty in the last two decimal places of the value.

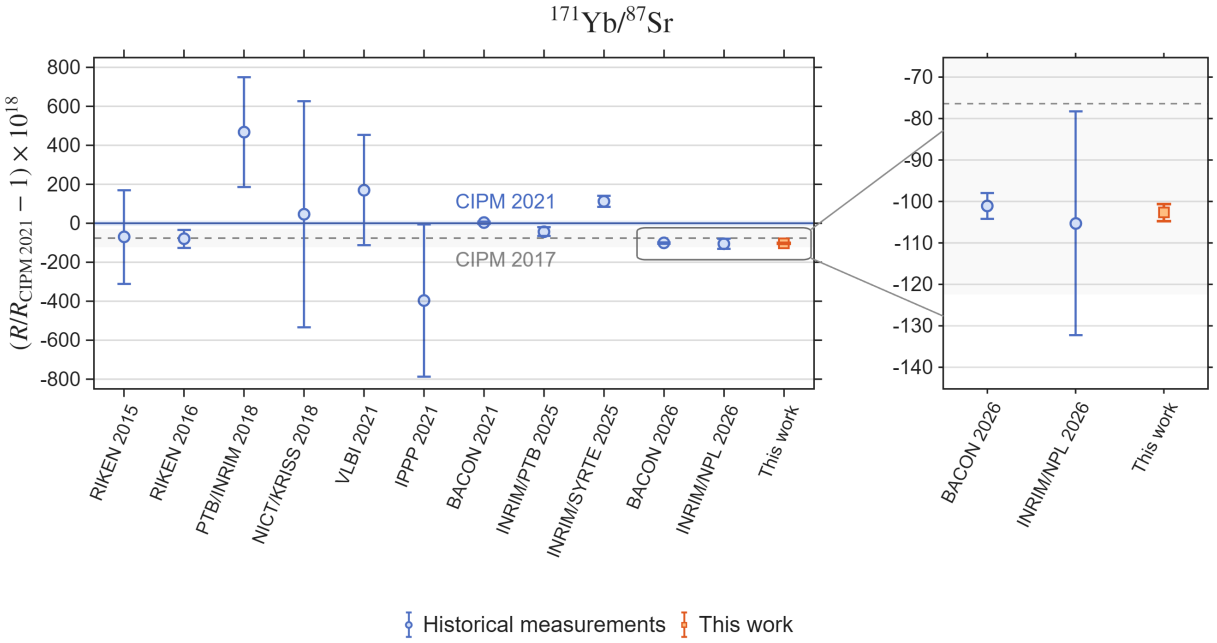


Figure 3: Evolution of the Yb/Sr frequency-ratio measurements. Deviations from the final value are shown against the measurement year for the historical determinations and for this work.

3 Discussion and outlook

The value reported here agrees with the NIST–JILA measurement at the level set by the redefinition roadmap (Figure 3): relative to the NIST–JILA value 1.207 507 039 343 337 723 0(37)[30] our central value differs by -2.87×10^{-18}, well within the 5×10^{-18} agreement threshold of the roadmap's frequency-ratio criterion[21;23], and within the combined standard uncertainty of the two measurements. The European-network value 1.207 507 039 343 337 718(32)[16], which differs from ours by $+2.13 \times 10^{-18}$, provides a supporting cross-check at a larger uncertainty of 3.2×10^{-17}.

The evolution of the Yb/Sr ratio also shows how comparison measurements drive the field forward. Since 2015[41], independent ratios were measured with increasing precision, yet the determinations disagreed well beyond their quoted uncertainties—an early sign of common-mode errors in the absolute frequency chain on which the earlier values relied[29]. The successive comparisons exposed those errors and guided their elimination (Figure 3).

Changes in the Earth's gravitational potential enter the comparison directly. The static difference between the two clock transitions is fixed by first-order spirit leveling and applied as a correction, although it rests on a single leveling campaign whose available GNSS-plus-geoid-model calibration is too imprecise (Methods) to act as an effective cross-check; closing this gap will require transportable optical clocks with higher frequency reproducibility, rather than geodetic techniques alone.

The time-varying part is observed directly. In any single segment the 1200-s demeaned beat waveform and the tidal template are too noisy to resolve, but across the 16 segments entering the ratio combination the combined evidence is decisive: the sign of the within-segment correlation is positive in 12 out of 16 segments for both tide models, and the precision-weighted response amplitude differs from zero at 5.0σ for the ETideLoad4.5 template (+0.58) and at 5.7σ for the DE421+FES2022 template (+0.87) (Methods; Supplementary Figure 1). A signal at the tidal period is therefore present in the record at the 10^{-18} level, yet no template captures it: the two amplitudes differ from each other and both fall short of the full expected response of +1, and the two models disagree in the sign of the center shift they produce. The observed coupling is smaller than the templates predict, so carrying the larger model shift as an uncertainty item

(Section 2.3) is conservative; a tide model that correctly describes the data would reduce this term, but constructing one requires more comparison data and dedicated geodetic inputs, which is beyond the scope of this paper.

The effect grows with the scale of the comparison: on longer, continental- and intercontinental-scale baselines the time-varying redshift becomes one of the dominant systematic terms[36]. That same sensitivity turns the network into an instrument for the Earth: clock comparisons over large baselines directly sense the time-variable gravitational potential, providing an independent means of testing and calibrating tide and loading models at the 10^{-18} level—the model disagreement seen here can be refined, or removed, by more comparison data from an expanding optical-clock network. Conversely, tidal knowledge gained in this way feeds back into the comparisons themselves: with data-consistent models, the time-varying term can be corrected rather than carried, extending the accuracy reach of large-scale clock comparisons.

Looking ahead, the result is a milestone on the necessary path toward a wide-area optical-clock network: the building blocks are now in place—terrestrial fiber links of the kind used here[13;14] carry comparisons over thousands of kilometers within continents, while free-space and satellite-based links are closing the remaining gaps[17–19;42]. The natural next step is a hybrid architecture in which optical-clock nodes on continental fiber backbones are interconnected across oceans by satellite or free-space optical links, so that any pair of clocks on Earth can be compared at the level demonstrated here. Such a network and the geodetic infrastructure will develop together: precise geopotential referencing supports the comparisons, while the network's time-variable gravity observations in turn refine the geodetic models[1;25].

4 Conclusion

We have compared an ytterbium lattice clock in Wuhan and a strontium lattice clock in Shanghai over a 1 350 km fiber link and determined the Yb/Sr frequency ratio with a total standard uncertainty of 2.34×10^{-18}. Together with the NIST–JILA measurement, this provides, for the first time, two independent determinations of the Yb/Sr ratio in agreement at the precision required by the roadmap for the redefinition of the SI second, satisfying criterion I.2.b of the roadmap for this pair of transitions. The two sites now form a working node pair of a distributed optical-clock network: with more optical-clock nodes joined through terrestrial fiber backbones and interconnected across continents by satellite and free-space links, the comparison demonstrated here extends naturally into the global network on which the redefinition, chronometric leveling of the gravitational potential, and distributed searches for new physics will depend.

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Methods

Leveling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the atomic-transition positions of the two clocks, determined by a first-order spirit leveling campaign

carried out specifically for this comparison. The leveling line, about 1 113 km long, runs between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC) through Ezhou, Huangshi, Huanggang, Lu'an, Hefei, Wuhu, Guangde and Suzhou. Digital levels were used with bar-code invar staffs, with sight lengths not exceeding 30 m; the line was observed in 6 km to 8 km sections in both directions, each checked in the field. Gravity was observed along the line at the leveling points with two relative gravimeters. The orthometric height difference between the benchmarks follows from the leveled normal heights after the normal-to-orthometric conversion, and the geopotential difference is integrated section by section, $\Delta W_{AB} = -\sum_i g_i \, \Delta H_i$, where g_i is the mean gravity and ΔH_i the orthometric height difference of the i-th section. Combining this with the two local offsets between benchmark and clock platform (obtained by trigonometric leveling) and the platform-to-transition offsets from the clock design parameters gives the atomic-transition geopotential difference $\Delta W = 280.042 \pm 0.085$ m^2 s^{-2}, equivalent to an optical fractional shift of -3.116×10^{-15}. The accidental error per kilometer of the leveling is $M_\Delta = 0.255\,\mathrm{mm\,km^{-1}}$, which satisfies the most stringent requirements for first-order leveling in the national standards of China, the United States, Germany and Czechia[43–46].

As an independent check, the geopotential difference was also determined by a combined GNSS and geoid-model calibration: the ellipsoidal heights of the permanent marks were measured by precise point positioning, their vertical offsets to the clock reference surfaces were leveled locally, and the EGM2008 geoid model was used to convert to a geopotential difference of 280.60 ± 2.558 m^2 s^{-2}. The two independent determinations agree within the uncertainties of the GNSS-based value, but the GNSS-plus-geoid calibration is two orders of magnitude less precise than the leveling result, so it cannot act as an effective cross-check at the reported accuracy; a meaningful independent verification will require transportable optical clocks with higher frequency reproducibility.

Experimental setup and frequency transfer

The comparison links two optical lattice clocks at different institutions: an ytterbium (Yb) clock at IAPMST in Wuhan and a strontium (Sr) clock at USTC in Shanghai, about 700 km apart. The two sites are connected by a phase-stabilized fiber link of 1 350 km single-pass length (2 700 km loop-back), operated at 1 550 nm. The route is divided into spans of tens of kilometers, each terminated by a relay station that carries a narrow-linewidth laser together with an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; the added instability of the link is kept below the clock noise level for averaging times beyond 100 s.

Both ends use independent optical frequency combs to bridge the spectral gap between the clock transitions and the telecom carrier. On the Shanghai side the Sr 698 nm transition is transferred via a 1 397 nm continuous laser to the 1 550 nm carrier; on the Wuhan side the received carrier is linked to the Yb 578 nm transition through the 1 156 nm light. All local frequency references and phase-locked loops are derived from the comb repetition rate $f_{\mathrm{rep}} = 200$ MHz. The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency $f_{1550} = 193.3992$ THz. All frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations, $f_{\mathrm{comb}}^{1397} = \frac{1}{2}(1+a)\, f_{\mathrm{Sr}}$ and $f_{\mathrm{comb}}^{1156} = \frac{1}{2}(1+b)\, f_{\mathrm{Yb}}$, where a and b collect the Sr and Yb systematic corrections (density, lattice, second-order Zeeman and blackbody shifts). The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor, $(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}} \times (1+\delta_g)$ with $\delta_g = -3.116 \times 10^{-15}$, derived from the leveled geopotential difference above.

Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Each accepted segment contributes its longest jump-free record. Supplementary Table 1 lists the per-segment retained counts, ratio deviations and statistical uncertainties; five segments whose stability slopes fall

outside the white-frequency-noise band are flagged there as model-limited. These five lie at the steep end of a continuous slope distribution, just beyond the acceptance bound, and show no common campaign, weather or laboratory condition that distinguishes them from the remaining segments.

Clock-ratio determination, statistical model and uncertainty budget

Each retained segment yields a clock-ratio estimate R_i from the mean of its retained beat samples, inverted through the fixed frequency-chain relations and corrected for the static gravitational redshift $\delta_g = -3.116 \times 10^{-15}$ derived from the leveled geopotential difference (see above). The per-segment statistical uncertainty u_i is obtained from the overlapping Allan deviation of the beat fractional frequency, evaluated on a tau grid of powers of two plus 600, 1200, 3600 and 7200 s, all limited to a quarter of the longest gap-free run T_i of the segment: u_i adopts the last computable point $u_i = R_i\,\sigma_y(\tau_{\max})$ without extrapolating the stability curve beyond the data. The log-log slope of σ_y against τ is computed as a diagnostic; segments whose slope leaves the white-frequency-noise band are flagged but retained.

The segment estimates are then combined under a random-effect model with a common excess scatter[32;33]: each segment is modeled as $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$, with ξ added in quadrature to the per-segment uncertainties. The primary value is obtained from a Bayesian random-effect model, in which μ and ξ are jointly inferred with flat priors on μ and on $\log\xi$, and the joint posterior $p(\mu, \xi \mid \{R_i, u_i\})$ is marginalized numerically over a 40-decade $\log\xi$ grid, with the grid span expanded until the posterior mass at the edges falls below 1%; the posterior mean of μ and its posterior standard deviation give the reported central value and statistical uncertainty, with $\xi = 0.14 \times 10^{-18}$. As cross-checks, the same observation model is combined by weighted least squares (inverse-variance weights $w_i = 1/u_i^2$) and by the Mandel–Paule estimator[33]; the resulting χ^2_{red} and ξ values are given in Table 1.

The remaining terms of the uncertainty budget (Table 2) are systematic. The Sr and Yb terms are the published evaluations of the two clocks[5;8]; the value adopted for the Yb clock resolves an internal-versus-published discrepancy in favor of the published evaluation. The static-potential term is $u(\Delta W)/c^2$ from the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$, giving 0.946×10^{-18}. The frequency-conversion stages contribute 0.1×10^{-18} and 0.1×10^{-18} from their frequency-reference settings.

Tidal template and time-varying redshift

The time-varying gravitational-redshift term is quantified from a tidal template. The supplied tide series consists of the solid Earth tide and ocean tide loading displacement difference between the two sites, computed every 30 s with the public software ETideLoad4.5[37], which implements the geophysical models and numerical standards of the IERS Conventions (2010)[38]. The displacement difference is converted to an equivalent geopotential difference with the local gravity g and then to a beat-frequency template through $h = f_{1550}\,\Delta W/c^2$, the same normalization used for the beat record.

To test how well the template fits the data, we repeated the ratio combination with the tide series applied before the inversion, at two fixed response amplitudes: the theoretical amplitude $A = +1$ and the historical amplitude $A = +0.54$ obtained from the demeaned beat waveform in our earlier analysis of the same campaign. The fit degrades rather than improves: the reduced chi-square of the combination rises from 0.41 to 0.57 and 0.54, respectively, and the Bayesian center moves by -0.85×10^{-18}. A second, independent tide template was evaluated as a cross-check: the solid tide was computed from the DE421 ephemerides (Earth–Sun–Moon positions) with Love numbers, and the ocean-tide loading correction to the gravitational potential was taken directly from the FES2022 global ocean-tide model, on a 90 s Beijing-time grid. Its fixed-amplitude combination again degrades the fit ($\chi^2_{\mathrm{red}} = 0.50$ at the historical amplitude) but moves

the Bayesian center in the opposite direction, by $+0.21 \times 10^{-18}$. To separate the solid-tide and ocean-loading contributions, we also repeated the segment-mean correlation with the solid tide alone: the Pearson correlation of the ratio deviations with the segment-mean full tide is +0.43, which falls to +0.36 for the solid tide only, indicating that the dominant coupling comes from the ocean-tide loading term.

Templates that degrade the segment agreement, and whose two independent realizations disagree in sign, are treated as inconsistent with the data: no tidal correction is applied to the reported central value, and the larger of the two center shifts is carried as the time-varying-redshift uncertainty item 0.85×10^{-18}.

Within-segment tidal correlation analysis

The within-segment correlation analysis uses the same 1200-s triangular window as the ratio processing: the retained beat record of each segment and the 1-s tidal template are both window-averaged with a 1200-s triangular window at 600-s stride, demeaned, and fitted through the origin as beat $= A \cdot$ tide, with the template sign corrected for the beat polarity so that a positive amplitude corresponds to the expected response $A = +1$; the amplitude, the Pearson correlation r and its p value are inferred from the non-overlapping 1200-s windows only, with the per-segment amplitude uncertainty obtained from the scatter of the non-overlapping residuals. Supplementary Figure 1a shows this fit for one segment: the template varies at the 10^{-17} level while the single-segment beat residuals are comparable in size, so no single segment reaches significance on its own. The same fit is repeated for both tide models, on the same 16 segments that enter the ratio combination (Supplementary Figure 1b). The per-segment amplitudes are combined with precision weights $1/u_A^2$; the sign consistency is evaluated by a two-sided binomial test on the number of segments with positive correlation, together with a signed Stouffer combination of the per-segment p values. The combined correlation is significant for both models, but neither model reproduces the expected response amplitude +1 (the observed precision-weighted amplitudes are +0.58 and +0.87), and the two models disagree with each other, so the observed coupling is smaller than the templates predict and the tidal term is kept as an uncertainty rather than a correction, as in the ratio combination above.

The components are treated as independent and combined in quadrature, giving the total standard uncertainty 2.34×10^{-18}.

Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (`https://github.com/hjiang555-a11y/gravitation-whsh`) under `results/audit-v1/verified/`: the manifest with SHA-256 `ac725aaba8ec744bf4e7a17cf37cee41af57461f0d3684584711b24a02b0962e`, the derived ledgers and statistics, and the generated audit report. The raw beat files and the leveled geopotential inputs are available from the corresponding author upon reasonable request.

Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.